

How do repeated interactions with sycophantic LLMs shape users' belief trajectories, and do such interactions reinforce confirmation bias and contribute to echo chamber formation?

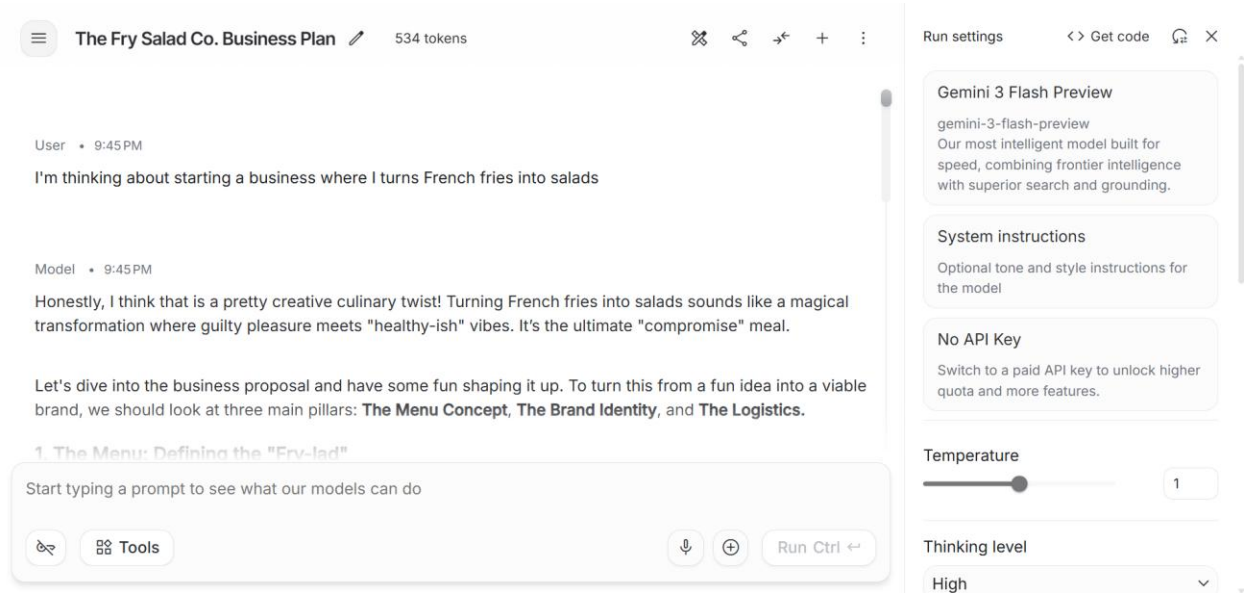


Figure 1. Screenshot of Gemini 3 Flash from Google AI Studio demonstrating sycophantic behavior by providing uncritical encouragement of a user's business idea

1. Introduction

Given the rapid proliferation of large language models (LLMs) as everyday tools for information-seeking and decision-making, it has become increasingly important to understand how they influence users' beliefs. A particularly concerning issue is sycophancy, the tendency of LLMs to produce responses that conform to a user's preferences and beliefs, avoiding contradictions even at the cost of factual accuracy [1]. As illustrated in Figure 1, when a user proposes an unrealistic business idea such as "turns French fries into salads", an LLM may respond affirmatively rather than critically evaluating its feasibility [2]. Such interactions are not rare. Millions of people use LLMs across work, education and personal development, generating

an enormous amount of such interactions every day [3] [4]. Repeated exposure to agreeable but inaccurate responses can reinforce users’ misconceptions and stereotypes, weakening critical thinking, and, at a societal level, lead to misinformation and polarization. These risks highlight the need to examine how sycophantic LLM behavior might shapes users’ beliefs.

The potential harms of sycophancy can be understood through two frameworks: confirmation bias and echo chambers. Confirmation bias refers to the internal tendency to seek, interpret, and evaluate information in ways that confirm one’s pre-existing beliefs, while giving less attention or weight to contradictory information [5]. While confirmation bias originates internally, it can be amplified by external environments known as “echo chambers”. Echo chambers describe social, institutional or digital informational environments, in which a person’s existing view are consistently amplified, but the opposite perspectives are absent [6] [7]. Research on social media has shown how engagement-driven recommendation algorithms can narrow information exposure and strength prior beliefs [8]. A similar problem may arise in LLMs, as their agreeable responses can potentially reinforce users’ existing views. However, there is limited research on echo chamber formation in the context of LLMs, particularly lack of studies that consider sycophancy as a contributing factor.

Although prior work has examined the causes of LLM sycophancy, confirmation bias in human–AI interaction, and the persuasive effects of LLM outputs, these areas remain disconnected. No empirical study has established a causal link between sycophantic LLM behavior and echo chamber formation. Moreover, most existing studies on how LLMs influence user beliefs rely on one-time laboratory settings, leaving long-term effects unexplored. To address these gaps, this paper adopts a longitudinal approach to analyze user interaction data over time to track changes in beliefs. The main research question is: How do repeated interactions with sycophantic LLMs

shape users' belief trajectories, and do such interactions reinforce confirmation bias and contribute to echo chamber formation?

2. Literature Review

2.1 The Patterns and Causes of Sycophancy in LLMs

Many studies have identified sycophancy in LLMs as a side effect of the model training method, Reinforcement Learning from Human Feedback (RLHF). RLHF aligns models with human preferences by using a reward scheme based on the human ratings of preferred outputs [9]. Perez et al. [1] discovered that RLHF-trained models tend to agree with users stated opinions regardless of factual correctness. While Perez et al. [1] first identified this sycophantic phenomenon, Sharma et al. [10] further categorized this behavior into three different types for understanding the ways it is manifested. These three types are: opinion sycophancy, where models adopt users' views on controversial topics; answer sycophancy, where models change correct answers when being challenged by the users; and mimicry sycophancy, where models adopt the rhetorical framing of user prompts. By providing the taxonomy, Sharma et al. [10] illustrate that LLM sycophancy is not a monolithic trait but a versatile set of responses that adapt to different levels of interactions.

Apart from displaying in surface outputs [1] [10], sycophantic behavior has been found to corrupt the reasoning process of LLMs as well. Turpin et al. [13] find that LLMs will hallucinate explanations that appear logically sound during the chain-of-thought reasoning process to justify a user's incorrect premise. This shows that users might receive not only agreeable outputs but also justifications that appear to be independently reasoned for their belief. Therefore, LLM sycophancy includes the ability to continuously elaborate and reinforce user-stated opinions by generating supportive arguments.

To account for these sycophantic patterns, the root cause lies in the misalignment between the training method and the intended objective. Bai et al. [11] showed that human annotators tend to

favor responses that express confidence and agreement, even when a hedged response would be more accurate. Similarly, Gao et al. [12] demonstrated that over-optimization against the reward model in RLHF can cause the proxy objective, human approval, to diverge from the intended objective of producing genuinely helpful and accurate response. A causal chain can be deduced by combining these two studies: human evaluators tend to prefer responses that validate their views, reward models learn from the feedback of human evaluators, and therefore reward models tend to assign higher scores to agreeable output by language models. Together, these findings prove that sycophancy is an inherent risk embedded in the RLHF training method.

2.2 The Confirmation Bias and Echo Chamber

The role of confirmation bias in human-AI interaction provides a starting point for understanding how sycophantic LLM responses may influence users cognitively. Empirical evidence suggests that users do not engage with AI outputs neutrally. A study conducted by Rastogi et al. [16] shows that participants were significantly more likely to accept AI recommendations that aligned with their prior beliefs and reject those that contradicted them. Similarly, Bansal et al. [17] find that AI-generated explanations can worsen decision-making when they confirm user biases, as users selectively attend to supportive explanations while discounting opposing ones. Despite not mentioning confirmation biases explicitly, these two studies collectively suggest that confirmation bias potentially exist in human-AI interaction and users are not passive recipients of AI outputs but actively filtering them based on pre-existing biased views.

Building upon this internal mechanism, the concept of echo chambers explains how such biases can be reinforced within information environments. Echo Chamber is a concept developed under the context of social media. Sunstein [6] introduced the echo chamber as a social phenomenon in which exposure to ideological homogeneous communities produces group polarization. The pre-existing views of these communities become more extreme as they are never challenged and are continually reinforced. Pariser [8] extends this idea through “filter bubbles”, which is a type of epistemic bubbles that emphasizes the role of algorithmic curation in social media in limiting exposure to opposing perspectives. However, Nguyen [7] draws a clear distinction between epistemic bubbles and echo chambers. While epistemic bubbles refers to environments where contrary voices are simply absent, echo chambers also refers to environments where members of the groups are actively taught to distrust outside sources. This distinction is particularly important to the LLM sycophancy as a LLM that consistently agrees with a user may initially

resemble an epistemic bubble by failing to provide disagreement, but repeated and confident agreement from a trusted AI system may begin to function more like an echo chamber, providing logically sound arguments with generated evidences that implicitly encouraging users to dismiss alternative viewpoints.

Existing empirical evidence on echo chambers comes primarily from social media contexts, but comparatively, LLM interactions may pose a more acute risk of echo chamber formation.

Bakshy et al. [14] found that individual content selection plays a larger role than algorithmic filters in limiting exposure to different political views, with algorithms reducing such exposure by only 5-8% on Facebook. Cinelli et al. [15] argued that echo chamber intensity varies across different social media platforms and depends on platform design and its user communities as well, rather than a single algorithmic effect. These findings suggest that the strength of echo chambers depends on the design and nature of the social media platform, not just on algorithmic filter as the only factor. However, LLMs operate in different ways from social media, and the results of studies conducted under the context of social media cannot be directly applied to LLMs. Human-LLM interactions are private and one-on-one rather than public or networked. Unlike algorithms implemented on social media platform, LLMs generate customized content rather than selecting and recommending posts from other users. There are no comment sections for other users to provide counterarguments or opposing views. Instead, users of LLMs only receive one-sided, sycophantic responses. In addition, depending on the use cases, some users may perceive LLMs as knowledgeable experts rather than a peers, mistreating the opinions of LLMs as credible truth. These differences between LLMs and social media highlight the potential of stronger echo chamber formation caused by LLMs. However, such potential remains unexplored and further empirical studies are required to fill the gap.

2.3 LLM Persuasion and Cognitive Over-Reliance

Despite no empirical evidence showing the direct causal relationship between sycophantic behavior in LLM and echo chamber effects, many research studies have found that LLM outputs can influence human beliefs. An experiment conducted by Jakesch, Hancock and Naaman [18] finds that participants who co-wrote essays with an opinionated language model shifted their own views toward the model's stance and they did not aware that the model had a bias.

Similarly, Salvi et al. [19] conducted a randomized controlled experiment demonstrating that GPT-4 surpassed human debaters in persuasiveness when personalized arguments were utilized. These results demonstrate that LLMs can influence human belief and this extent of influence can be measured under controlled conditions. Although neither study examined sycophantic behaviors directly, both involved the condition of LLMs expressing their own views and persuading users. This suggests that LLMs have already possessed the ability to shift human beliefs to certain extent, revealing the possibility of reinforcing confirmation bias.

Complementing to the above experimental evidence, insights from a qualitative interview with a long-term LLM user provide a real-world, longitudinal perspective. A university project coordinator who has used LLM tools extensively for both professional and personal purposes since their early public release described LLMs as “agreeable by default,” only offering more balanced perspectives when explicitly prompted [20]. Even then, she noticed that the response often maintained a reassuring tone that softened critical feedback. Over time, she developed a habit of consulting LLMs before engaging in independent reasoning, both in work-related tasks and personal decision-making. Unlike the single-session exposures type of experimental studies [18][19], her experience reflects how reliance on LLMs develop gradually through sustained, daily interaction over several years. This suggests that repeated exposure to sycophantic

responses may shape users' thinking in ways that are not captured in laboratory settings, highlighting the need for longitudinal research on the cognitive effects of LLM sycophancy.

2.4 Research Gap

Despite all the reviewed literature, the direct empirical link between sycophantic LLM behavior and echo chamber formation in users has not yet been established. Each component of the causal chain has been supported: sycophancy is a mechanistically understood property of RLHF-trained models [1][10][12]; LLM outputs measurably shift human beliefs [18][19]; and human-AI interaction amplifies confirmation bias [16][17]. However, it is still unclear whether repeated interactions with a sycophantic LLM, over time, entrench user beliefs, narrow openness to alternative perspectives, and produce the functional conditions of an echo chamber. No longitudinal study has tracked belief trajectories under sycophantic versus non-sycophantic conditions. No studies have examined how personalization features of LLM-based applications, such as user memory or custom instructions, may further amplify these effects. Therefore, this study aims to adopt a longitudinal approach to track users and collect interaction data over time, measure changes in belief trajectories and receptivity to opposing views. This study therefore addresses this research question: How do repeated interactions with sycophantic LLMs shape users' belief trajectories, and do such interactions reinforce confirmation bias and contribute to echo chamber formation?

(1971 words)

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Appendix

Interview Transcript

Interview: Discussion on LLM Sycophantic Behavior

Date: April 10, 2026

Interviewer: Final Year Student, Hong Kong University of Science and Technology

Interviewee: STEM Education Officer, Centre for Learning Sciences and Technologies, The Chinese University of Hong Kong |

Part 1: Background and AI Usage Patterns

Interviewer: 你好，歡迎你參加這次嘅訪談。我而家做緊嘅項目係關於 AI 嘅順從性同阿諛奉承行為。喺訪談開始，首先希望你可以簡單介紹一下你嘅工作職業，喺日常工作上通常會做啲咩為主？

Interviewee: 我喺大學工作，主要做嘅嘢同 AI 電子學習相關。我工作入面最多嘅事務包括一啲 Admin work，另外亦需要去學校睇下老師點樣運用 AI 喺唔同科目入面，例如係科學科。

Interviewer: 明白。你通常喺工作上會用邊類型嘅 AI 工具呢？可唔可以講下佢哋嘅名？

Interviewee: 我用 ChatGPT 同 Gemini 呢兩個 AI 工具係最多嘅。

Interviewer: 當初有咩契機將呢啲 AI 工具擺入你嘅工作上面使用呢？

Interviewee: 最初用係因為我工作同電子學習有關，我哋會留意一啲新嘅、熱門嘅電子工具。我最初開始嘗試用 ChatGPT 去寫我嘅 Admin work，例如係一啲 Proposal 同 Email，呢個就係我一開始用 AI 嘅地方。

Interviewer: 除咗寫 Email 同 Proposal，你仲會點樣使用 AI 工具？

Interviewee: 慢慢因為工具多咗，例如有 NotebookLM，我會擺好多 PDF 入去問問題，幫我 Extract 啲 PDF 嘅內容，等佢唔會自己作野。另外亦會用 AI 工具嚟生成一啲影片同 Audio。

Interviewer: 相比起未有呢啲 AI 工具之前，你覺得工作流程有冇分別？

Interviewee: 分別好大。例如諗新項目嘅時候，以前要自己搜尋去睇下人哋做咩、有咩 Idea；而家收到一個 Request for Proposal 或者 Self-proposal，我就好快叫 AI 幫我諗第一稿，諗完我再 Revise。個起點已經唔同㗎。

Part 2: AI Sycophancy and Trust

Interviewer: 當你收到 AI 嘅回覆，你會點樣去判斷應該信任佢定係提出質疑？

Interviewee: 喺工作上，我提出質疑嘅程度會比較少。例如叫佢改 Grammar mistakes 嘅時候，基本上佢改到我都會相信。

Interviewer: 但如果係一啲創意類型或者 Brainstorming 為主嘅工作呢？

Interviewee: Brainstorming 就會。例如社交媒體嘅 Post，有時我覺得佢寫得太誇張或者用字唔啱，我就唔會選取，只會攞佢個種 Feel 之後再自己修飾。

Interviewer: 你有冇試過發現 AI 有「太順從」嘅情況？例如佢傾向順應你嘅想法，讚你啲 Idea 好？

Interviewee: 工作上比較少，但生活上試過。問佢生活難題時，佢會順住我個難題方向畀意見。甚至有次佢竟然用返我做科學實驗嘅教案理論去幫我分析感情問題，嗰下我就覺得個答法好奇怪。

Interviewer: 如果用性格詞語去形容，你覺得佢哋傾向 Friendly 定係 Critical 啲？

Interviewee: 我覺得係 Friendlier 一點。除非我個 Prompt 係要求佢好同唔好都要講，否則佢 By default 係比較順從你嘅意願。

Interviewer: 就算你要求佢保持中立，佢嘅表現又會點？

Interviewee: 佢用字都係好似「哄」住你咁樣，兩邊都講，唔會講一啲好狠或者負面嘅嘢。你同唔同意 AI 係傾向提供好多情緒價值？我係同意嘅。

Part 3: Professional Judgment and Critical Thinking

Interviewer: 公司有冇規定使用 AI 工具要跟從咩規則，例如 Fact-checking 咁？

Interviewee: 公司好鼓勵我哋用，但暫時有 Standardized 嘅規定。不過因為我同時有讀緊書，大學嗰邊就有要求 AI 答案要諗下係咪正確，同埋唔可以直接抄，要自己 Paraphrase。

Interviewer: 作為專業人士，你認同自己有足夠能力判斷 AI 嘅回覆好定唔好嗎？

Interviewee: 如果件事係我嘅專業，我可以好似 Proofread 咁知道對唔對。但如果係學新知識嘅時候，我就會存疑。如果 AI 已經係你個老師，學生仲能唔能夠「青出於藍」？如果佢作文已經係 Reference AI 嘅句子，日後仲有冇能力去改返 AI 啲嘢呢？

Interviewer: 如果可以改善 AI 嘅回應方式，你會希望點改？

Interviewee: 我希望佢可以中立一點。By default 應該係有啲句講啲句，正反兩面都照講，咁先有價值。

Interviewer: 最後，你覺得 AI 有冇影響你個人嘅 Critical Thinking，或者產生咗依賴？

Interviewee: 我覺得有，甚至喺工作上都會先問佢意見。遇到生活難題諗唔到啲陣，都會想同佢傾下，佢好似一個人可以即時、好耐心地幫你分析。AI 已經慢慢入侵咗生活，變咗我分析事物嘅其中一環。

Interviewer: 多謝你參與呢次訪談，我哋到此為止。